

Defination of Sets

- Sets are **well defined** collection of **objects or elements**.
- We write $a \in A$ to denote that a is an element of the set A .
- The notation $a \notin A$ denotes that a is not an element of the set A .
- It is common for sets to be denoted using uppercase letters. Lowercase letters are usually used to denote elements of sets.
- The set V of all vowels in the English alphabet can be written as

$$V = \{a, e, i, o, u\}.$$

Describing a set in such manner is known as the **roster method**.

1

Set Builder Notation

- We characterize all those elements in the set by stating the property or properties they must have to be members.
- For instance, the set O of all odd positive integers less than 10 can be written as

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

2

Numbers Sets

- $N = \{0, 1, 2, 3, \dots\}$, the set of **natural numbers**
- $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of **integers**
- $Z^+ = \{1, 2, 3, \dots\}$, the set of **positive integers**
- $Q = \{p/q \mid p \in Z, q \in Z, \text{ and } q \neq 0\}$, the set of **rational numbers**
- R , the set of **real numbers**
- R^+ , the set of **positive real numbers**
- C , the set of **complex numbers**.

3

Equal Sets

- Two sets are equal if and only if they have the same elements.
- Therefore, if A and B are sets, then A and B are equal if and only if

$$\forall x(x \in A \leftrightarrow x \in B).$$

Example

The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements.

Note :

- The order in which the elements of a set are listed does not matter.
- It does not matter if an element of a set is listed more than once, so $\{1, 3, 3, 3, 5, 5, 5, 5\}$ is the same as the set $\{1, 3, 5\}$ because they have the same elements

4

Null Set

- This set is called the empty set, or null set, and is denoted by $\emptyset$.
- The empty set can also be denoted by $\{ \}$.

Example

- The set of all positive integers that are greater than their squares is the null set.

5

Singleton Set

- A set with one element is called a singleton set.
- A common error is to confuse the empty set $\emptyset$ with the set $\{\emptyset\}$, which is a singleton set. The single element of the set $\{\emptyset\}$ is the empty set itself!
- A useful analogy for remembering this difference is to think of folders in a computer file system. The empty set can be thought of as an empty folder and the set consisting of just the empty set can be thought of as a folder with exactly one folder inside, namely, the empty folder.

6

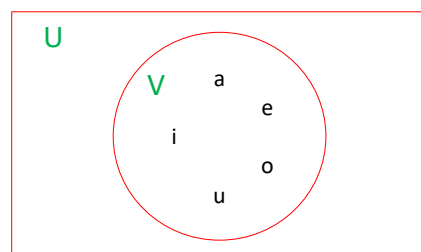
Venn Diagram

- Sets can be represented graphically using Venn diagrams,
- In Venn diagrams the universal set U , which contains all the objects under consideration, is represented by a rectangle.
- Inside this rectangle, circles or other geometrical figures are used to represent sets.
- Venn diagrams are often used to indicate the relationships between sets.

7

Venn Diagram

- We draw a rectangle to indicate the universal set U , which is the set of the 26 letters of the English alphabet. Inside this rectangle we draw a circle to represent V . Inside this circle we indicate the elements of V



8

Venn Diagram

- Draw the venn diagram for $(A - B) \cup (A - C) \cup (B - C)$

9

Subset

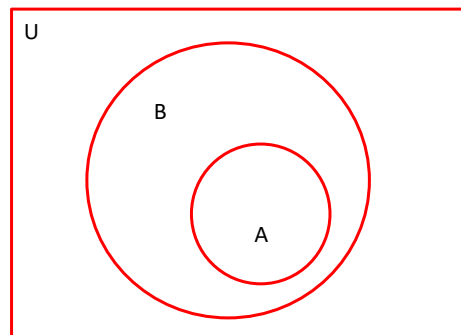
- The set A is a subset of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

$A \subseteq B$ if and only if the quantification $\forall x(x \in A \rightarrow x \in B)$

- To show $A \subseteq B$, show that if x belongs to A then x
- To show $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.

10

Venn Diagram Showing that A Is a Subset of B



11

For every set S , $\emptyset \subseteq S$.

- Let S be a set.
- To show that $\emptyset \subseteq S$, we must show that $\forall x(x \in \emptyset \rightarrow x \in S)$ is true.
- Because the empty set contains no elements, $x \in \emptyset$ is always false.
- It follows that the conditional statement $x \in \emptyset \rightarrow x \in S$ is always true, because its hypothesis is always false and
- a conditional statement with a false hypothesis is true.
- Therefore, $\forall x(x \in \emptyset \rightarrow x \in S)$ is true.

12

For every set S , $S \subseteq S$.

- Do it yourself ...

13

Proper Subsets

- When a set A is a subset of a set B but $A \neq B$, we write $A \subset B$ and say that A is a proper subset of B .
- For $A \subset B$ to be true, it must be the case that $A \subseteq B$ and there must exist an element x of B that is not an element of A . That is, A is a proper subset of B if and only if

$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$ is true.

14

Equal Sets

- To be equal sets, two sets must have the same elements i.e each set is a subset of the other.
- In other words, we can show that if A and B are sets with $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- That is, $A = B$ if and only if $\forall x(x \in A \rightarrow x \in B)$ and $\forall x(x \in B \rightarrow x \in A)$
or
- equivalently if and only if $\forall x(x \in A \leftrightarrow x \in B)$, which is what it means for the A and B to be equal.

15

Cardinality of Set

- Let S be a set. If there are exactly n distinct elements in S where n is a non-negative integer, we say that S is a finite set and that n is the cardinality of S. The cardinality of S is denoted by $|S|$.

Examples:

- Let A be the set of odd positive integers less than 10. Then $|A| = 5$.
- Because the null set has no elements, $|\emptyset| = 0$.

Note:

A set is said to be infinite if it is not finite.

- The set of positive integers is infinite.

16

Power sets

- Given a set S , the power set of S is the **set of all subsets** of the set S . The power set of S is denoted by $P(S)$.

- What is the power set of the set $\{0, 1, 2\}$?

The power set $P(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$.

$$P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

- What is the power set of the empty set? What is the power set of the set $\{\emptyset\}$?

The set $\{\emptyset\}$ has exactly two subsets, namely, $\emptyset$ and the set $\{\emptyset\}$ itself.

$$P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}.$$

17

Ordered Pairs

- The ordered n -tuple $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, $\dots$, and a_n as its n th element.
- We say that two ordered n -tuples are equal if and only if each corresponding pair of their elements is equal.
- In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2, \dots, n$.
- In particular, ordered 2-tuples are called ordered pairs. The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

18

Cartesian Product of Sets

- Let A and B be sets.
- The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

- What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

- Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$.

19

Cartesian product of more than two sets.

- The Cartesian product of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$, where a_i belongs to A_i for $i = 1, 2, \dots, n$. In other words,

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}.$$

- What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?

Solution: The Cartesian product $A \times B \times C$ consists of all ordered triples (a, b, c) , where $a \in A$, $b \in B$, and $c \in C$. Hence,

$$A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}.$$

20

Cartesian product of itself

- A^2 to denote $A \times A$, the Cartesian product of the set A with itself.
- $A^3 = A \times A \times A$,
- $A^4 = A \times A \times A \times A$, and so on.
- $A^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A \text{ for } i = 1, 2, \dots, n\}$.

Suppose that $A = \{1, 2\}$.

$$A^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

$$A^3 = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}.$$

21

Set Operation

- Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.
 - $A \cap B$
 - $A \cup B$
 - $A - B$
 - $B - A$
- Suppose that A is the set of freshers at your college and B is the set of students in software engineering at your college. Express each of these sets in terms of A and B .
 - the set of freshers taking software engineering in your college
 - the set of freshers at your college who are not taking software engineering
 - the set of students at your college who either are freshers or are taking software engineering
 - the set of students at your school who either are not freshers or are not taking software engineering

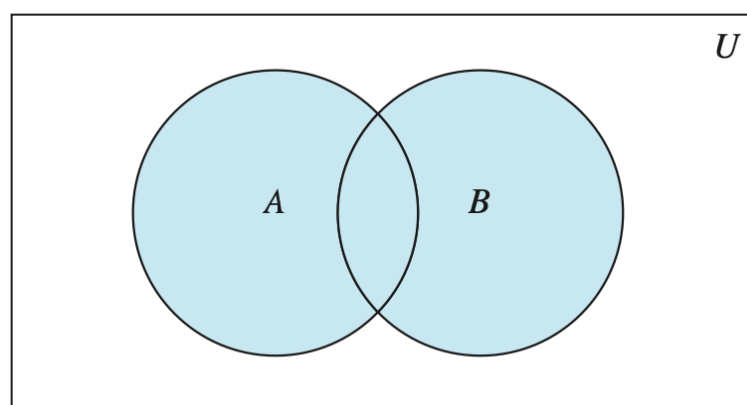
22

Set Operations – Union

- Let A and B be sets.
- The union of the sets A and B , denoted by $A \cup B$, is the set that **contains those elements that are either in A or in B , or in both.**
- An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that
$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$
- The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$.

23

Set Operations – Union



$A \cup B$ is shaded.

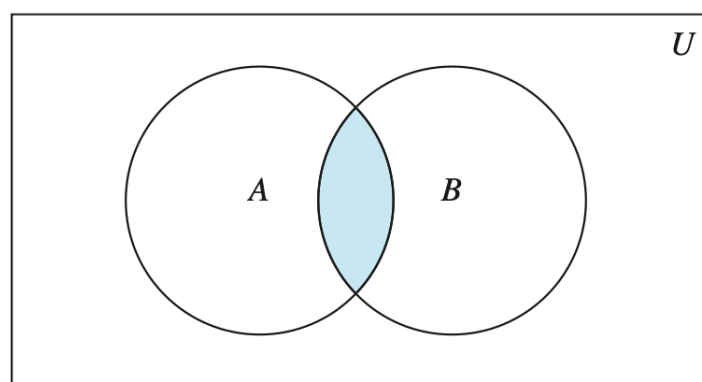
24

Set Operations – Intersection

- Let A and B be sets.
- The intersection of the sets A and B , denoted by $A \cap B$, is the set **containing those elements in both A and B .**
- An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B . This tells us that
$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$
- **The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 3\}$; that is, $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$.**

25

Set Operations – Intersection



$A \cap B$ is shaded.

26

Set Operation – Difference

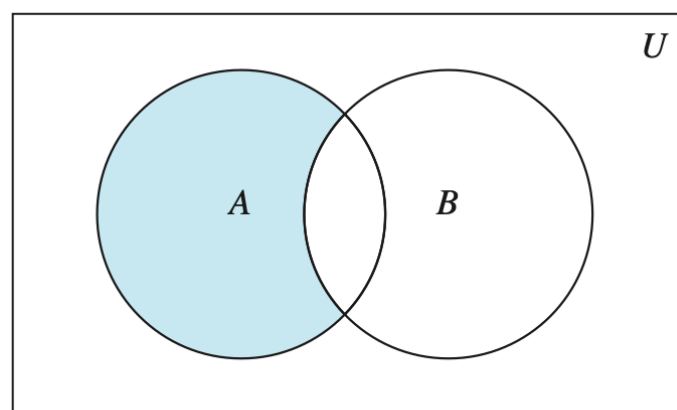
- Let A and B be sets.
- The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the complement of B with respect to A .
- An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that

$$A - B = \{x \mid x \in A \wedge x \notin B\}.$$

- The difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{5\}$; that is, $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$. This is different from the difference of $\{1, 2, 3\}$ and $\{1, 3, 5\}$, which is the set $\{2\}$.

27

Set Operation – Difference



$A - B$ is shaded.

28

Set Operation – Complement

- Let U be the universal set. The complement of the set A , denoted by $\bar{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.
- An element belongs to A if and only if $x \in A$. This tells us that

$$A = \{x \in U \mid x \in A\}.$$
- Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then

$$\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}.$$

29

Set Identities

<i>Identity</i>	<i>Name</i>
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\bar{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \bar{A} \cup \bar{B}$ $\overline{A \cup B} = \bar{A} \cap \bar{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

30

De-Morgan's Law

Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
 \overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{by definition of complement} \\
 &= \{x \mid \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol} \\
 &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
 &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan law for logical equivalences} \\
 &= \{x \mid x \notin A \vee x \notin B\} && \text{by definition of does not belong symbol} \\
 &= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{by definition of complement} \\
 &= \{x \mid x \in \overline{A} \cup \overline{B}\} && \text{by definition of union} \\
 &= \overline{A} \cup \overline{B} && \text{by meaning of set builder notation}
 \end{aligned}$$

31

Membership Table

- Set identities can also be proved using membership tables.
- We consider each combination of sets that an element can belong to and verify that elements in the same combinations of sets belong to both the sets in the identity.
- To indicate that an element is in a set, a 1 is used; to indicate that an element is not in a set, a 0 is used.

32

Membership Table

- Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

33

Generalized Union and Intersections

- The union of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

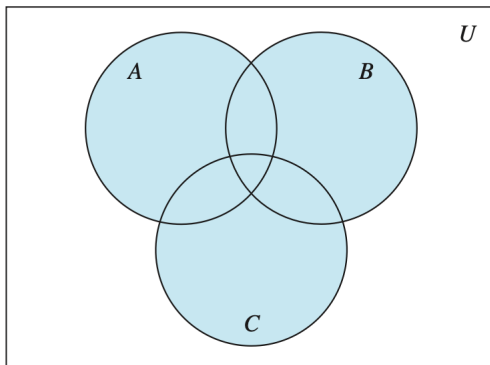
$$A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

- The intersection of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

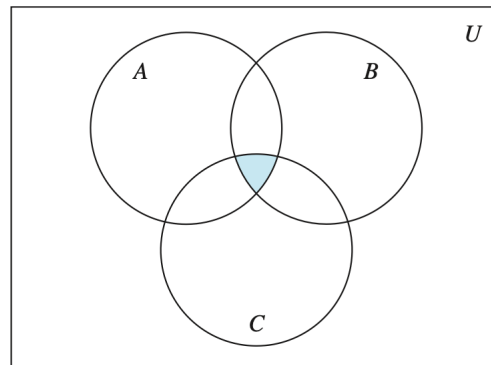
$$A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

34

Generalized Union and Intersections



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

35

Principle of Inclusion and Exclusion

- We are often interested in finding the cardinality of a union of two finite sets A and B . that $|A| + |B|$ counts each element that is in A but not in B or in B but not in A exactly once, and each element that is in both A and B exactly twice.
- Thus, if the number of elements that are in both A and B is subtracted from $|A| + |B|$, elements in $A \cap B$ will be counted only once.

$$\text{Hence, } |A \cup B| = |A| + |B| - |A \cap B|.$$

- The generalization of this result to unions of an arbitrary number of sets is called the principle of inclusion–exclusion.

- PIE for 3 sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

36

Disjoint Sets

- Two sets are called disjoint if their intersection is the empty set.
- Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Because $A \cap B = \emptyset$, A and B are disjoint.

37

Computer Representation of Sets

- Assume that the universal set U is finite (and of reasonable size so that the number of elements of U is not larger than the memory size of the computer being used).
- First, specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$.
- Represent a subset A of U with the bit string of length n , where the i^{th} bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A .
- Using bit strings to represent sets, it is easy to find complements of sets and unions, intersections, and differences of sets.

38

Computer Representation of Sets

- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order; that is, $a_i = i$. What bit strings represent the subset of all odd integers in U , the subset of all even integers in U , and the subset of integers not exceeding 5 in U ?
- **Solution:** The bit string that represents the set of odd integers in U , namely, $\{1, 3, 5, 7, 9\}$, has a one bit in the first, third, fifth, seventh, and ninth positions, and a zero elsewhere. It is 1010101010.
- Similarly, we represent the subset of all even integers in U , namely, $\{2, 4, 6, 8, 10\}$, by the string 0101010101.
- The set of all integers in U that do not exceed 5, namely, $\{1, 2, 3, 4, 5\}$, is represented by the string 1111100000.

39

Bit Stream representation of Sets - Exercise

- We have seen that the bit string for the set $\{1, 3, 5, 7, 9\}$ (with universal set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$) is 10 1010 1010. What is the bit string for the complement of this set?
- The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

40

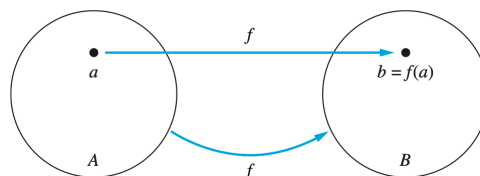
Read about these and present next class ...

1. Multisets
2. Fuzzy Sets

41

Functions

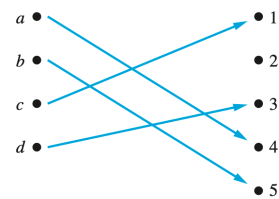
- Let A and B be nonempty sets. A function f from A to B is an assignment of exactly one element of B to each element of A .
- We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .
- If f is a function from A to B , we write $f : A \rightarrow B$.



42

One-to-One Function

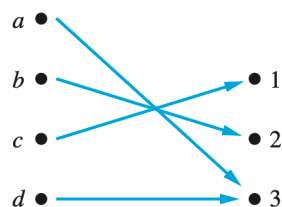
- A function f is said to be one-to-one, or an injection, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be injective if it is one-to-one.
- f is one-to-one using quantifiers as $\forall a \forall b (f(a) = f(b) \rightarrow a = b)$ or equivalently $\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$



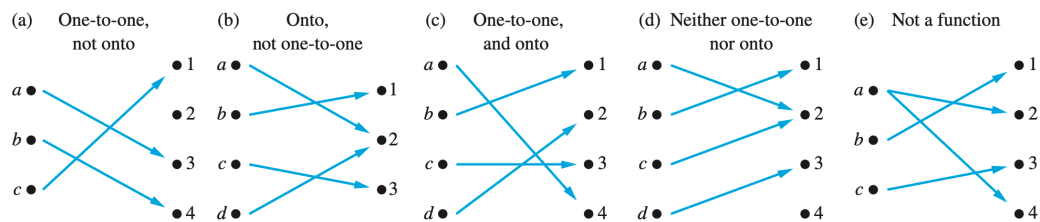
43

Onto Function

- A function f from A to B is called onto, or a surjection, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called surjective if it is onto.
- f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$.



44



45

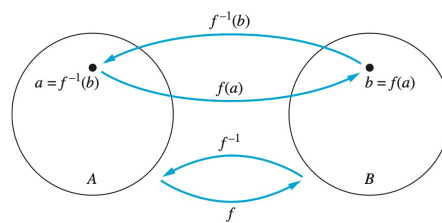
Bijjective Function

- The function f is a one-to-one correspondence, or a bijection, if it is both one-to-one and onto. We also say that such a function is bijective.
- Example, f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a)= 4$, $f(b)= 2$, $f(c)= 1$, and $f(d)= 3$.

46

Inverse of a function

- Let f be a one-to-one correspondence from the set A to the set B .
- The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$.
- The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.



47

Inverse of a function

$$f(x) = x + 1$$

$$y = x + 1$$

$$x = y - 1$$

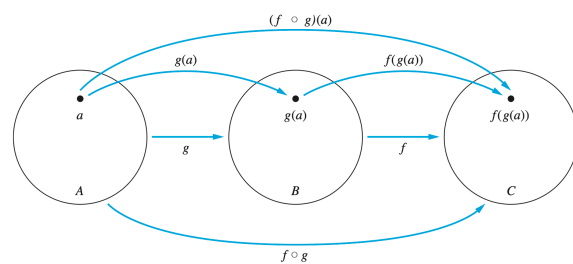
$$f^{-1}(y) = y - 1$$

48

Composit Functions

- Let g be a function from the set A to the set B and let f be a function from the set B to the set C .
- The composition of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$



49

Composit function

- If $f(x) = 2x + 3$ and $g(x) = 3x + 2$
- $f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$
- $g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$

50

The Graphs of Functions

- Let f be a function from the set A to the set B . The graph of the function f is the set of ordered pairs $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.
- Example: The graph of f is the set of ordered pairs of the form $(x, f(x)) = (x, x^2)$, where x is an integer
- The graph of f is the set of ordered pairs of the form $(x, f(x)) = (x, x^2)$, where x is an integer.

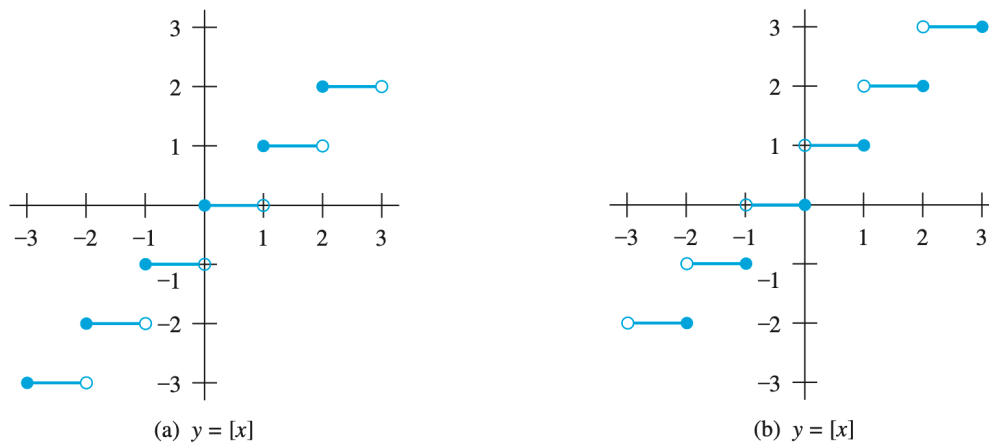
51

Floor and Ceiling Function

- The floor function assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$. The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.

$$\lfloor \frac{1}{2} \rfloor = 0, \lceil \frac{1}{2} \rceil = 1, \lfloor -\frac{1}{2} \rfloor = -1, \lceil -\frac{1}{2} \rceil = 0, \lfloor 3.1 \rfloor = 3, \lceil 3.1 \rceil = 4, \lfloor 7 \rfloor = 7, \lceil 7 \rceil = 7.$$

52



Graphs of the (a) Floor and (b) Ceiling Functions.

53

Use of Floor and Ceiling Function

- Each byte is made up of 8 bits. How many bytes are required to encode 100 bits of data?
- Solution:
- To determine the number of bytes needed, we determine the smallest integer that is at least as large as the quotient when 100 is divided by 8, the number of bits in a byte.
- Consequently, $\lceil 100/8 \rceil = \lceil 12.5 \rceil = 13$ bytes are required.

54

Boolean Functions

- defines a logical relationship between the binary variables and binary output.
- Binary variable has value 0 or 1
- Logical Operators may be AND, OR, NOT
- Truth tables are used to represent the Boolean Functions
- K-Maps are used to simplify the Boolean functions
- Example $Y(A,B) = A'B + B'A$

55

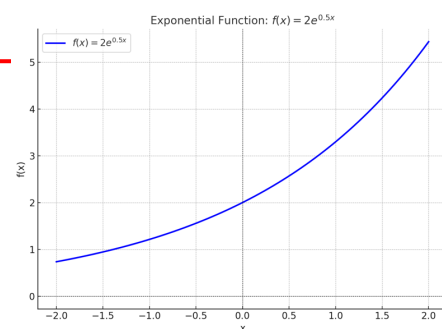
Exponential Functions

- The exponential function is defined as:

$$f(x) = a \cdot e^{bx}$$

Where:

- a is the initial value (amplitude),
- e is Euler's number (≈ 2.718),
- b determines the growth/decay rate.
- Rapid increase (for $b > 0$) or rapid decrease (for $b < 0$).
- Example: $f(x) = 2e^{0.5x}$



56

Division in Integers

If $n \neq 0$ and m are nonnegative integers, we can write $m = qn + r$ for some nonnegative integers q and r with $0 \leq r < n$.

Example:

- a) If n is 3 and m is 16, then $16 = 5(3) + 1$, so q is 5 and r is 1.
- b) If n is 10 and m is 3, then $3 = 0(10) + 3$, so q is 0 and r is 3.

57

Division in Integers

Let a, b and c be integers.

- (a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- (b) If $a \mid b$ and $a \mid c$, where $b > c$, then $a \mid (b - c)$.
- (c) If $a \mid b$ or $a \mid c$, then $a \mid bc$.
- (d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Verify these statements with suitable examples.

Note: $\mid$ denotes Divisibility: if m and n are two integers, $m \mid n$ means that m divides n evenly.

58

Prime Numbers

A number $p > 1$ in $\mathbb{Z}^*$ is called prime if the only positive integers that divide p are p and

The numbers 2, 3, 5, 7, 11, and 13 are prime, while 4, 10, 16, and 21 are not prime.

59

Prime Numbers

Algorithm to test whether an integer $N > 1$ is prime:

- STEP 1. Check whether N is 2. If so, N is prime. If not, proceed to
- STEP 2. Check whether $2 \mid N$. If so, N is not prime; otherwise, proceed to
- STEP 3. Compute the smallest integer $K \leq \sqrt{N}$. Then
- STEP 4. Check whether $D \mid N$, where D is any odd number such that $1 < D \leq K$.

If $D \mid N$, then N is not prime; otherwise, N is prime.

60

Greatest Common Divisor

- If a , b , and k are in $\mathbb{Z}^*$, and $k|a$, $k|b$, we say that k is a common divisor of a and b .
- If d is the largest such k , d is called the greatest common divisor, or GCD, of a and b , and we write $d = \text{GCD}(a,b)$.
- If $\text{GCD}(a, b) = 1$, we say that a and b are relatively prime.

61

Properties of GCD

- If d is $\text{GCD}(a, b)$, then
 - $d = sa + tb$ for some integers s and t (these are not both positive).
 - If c is any other common divisor of a and b , then $c|d$.

Example:

The common divisors of 12 and 30 are 2, 3, and 6, so $\text{GCD}(12, 30) = 6$ and $6 = 1 \cdot 30 + (-2) \cdot 12$.

62

Eucliden Algorithm to find GCD

- To find GCD we need four variables namely,
 - '**q**' for quotient
 - '**r1**' for greater number in given pair
 - '**r2**' for smaller number in given pair and
 - '**r**' for remainder
- We divide r1 by r2 and note values of q and r in tabular form
- Shift r1, r2 and r for each iteration untill the division is possible
- The final r1 value after all iteration will be GCD of two numbers.

63

Eucliden Algorithm to find GCD Example

- Find GCD of (32 and 86) by using Eucliden algorithm.

Quotient(q)	r1	r2	Remainder(r)
2	86	32	22
1	32	22	10
2	22	10	2
5	10	2	0
X	2	0	X

64

Extended Eucliden Algorithm to find GCD

- This is more like Eucliden Algorithm but we add some new parameters to calculate GCD.
- To find GCD we need,
 - '**q**' for quotient
 - '**r1**' for greater number in given pair
 - '**r2**' for smaller number in given pair and
 - '**r**' for remainder
 - '**s1**' = 1
 - '**s2**' = 0
 - '**s**' = $s1 - s2 * q$
 - '**t1**' = 0
 - '**t2**' = 1
 - '**t**' = $t1 - t2 * q$

65

Eucliden Algorithm to find GCD Example

- Find GCD of (26 and 68) by using Eucliden algorithm.

q	r1	r2	r	s1	s2	s	t1	t2	t
3	68	26	8	1	0		0	1	
	26	8							

66

Matrices

- A matrix is a rectangular array of numbers arranged in **m** horizontal rows and **n** vertical columns:
- The following matrix gives the airline distances between the cities indicated.

	London	Madrid	New York	Tokyo
London	0	785	3469	5959
Madrid	785	0	3593	6706
New York	3469	3593	0	6757
Tokyo	5959	6706	6757	0

67

Properties of matrices

- Two $m \times n$ matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if $a_{ij} = b_{ij}$, $1 \leq i \leq m$, $1 \leq j \leq n$, that is, if corresponding elements are the same.
- If $A = [a_{ij}]$ and $B = [b_{ij}]$ are $m \times n$ matrices, then the sum of A and B is the matrix $C = [c_{ij}]$ defined by $c_{ij} = a_{ij} + b_{ij}$, $1 \leq i \leq m$, $1 \leq j \leq n$. that is, C is defined by adding the corresponding elements of A and B.
- Properties of matrix addition:
 - $A + B = B + A$.
 - $(A + B) + C = A + (B + C)$.
 - $A + 0 = 0 + A = A$.

68

Properties of Matrices

- If $A = [a_{ij}]$ is an $m \times p$ matrix and $B = [b_{ij}]$ is a $p \times n$ matrix, then the product of A and B , denoted AB , is the $m \times n$ matrix $C = [c_{ij}]$ defined by

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n$$
- Properties of Matrix multiplication:
 - $A(BC) = (AB)C$.
 - $A(B + C) = AB + AC$.
 - $(A + B)C = AC + BC$.

69

Properties of Matrices

If $\mathbf{A} = [a_{ij}]$ is an $m \times n$ matrix, then the $n \times m$ matrix $\mathbf{A}^T = [a_{ij}^T]$, where $a_{ij}^T = a_{ji}$, $1 \leq i \leq m$, $1 \leq j \leq n$, is called the **transpose of $\mathbf{A}$** . Thus the transpose of $\mathbf{A}$ is obtained by interchanging the rows and columns of $\mathbf{A}$.

Theorem 3. If $\mathbf{A}$ and $\mathbf{B}$ are matrices, then

- $(\mathbf{A}^T)^T = \mathbf{A}$.
- $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$.
- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.

70

Boolean Matrix

- A boolean matrix is an $m \times n$ matrix whose entities are either zero or one.

Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ Boolean matrices. We define $\mathbf{A} \vee \mathbf{B} = \mathbf{C} = [c_{ij}]$, the **join** of $\mathbf{A}$ and $\mathbf{B}$, by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both 0.} \end{cases}$$

and $\mathbf{A} \wedge \mathbf{B} = \mathbf{D} = [d_{ij}]$, the **meet** of $\mathbf{A}$ and $\mathbf{B}$, by

$$d_{ij} = \begin{cases} 1 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both 1} \\ 0 & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0. \end{cases}$$

71

Properties of Boolean Matrices

- $\mathbf{A} = [a_{ik}]$ is an $m \times p$ Boolean matrix and $\mathbf{B} = [b_{kj}]$ is a $p \times n$ Boolean matrix.

The **Boolean product** of $\mathbf{A}$ and $\mathbf{B}$, denoted $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ Boolean matrix $\mathbf{C} = [c_{ij}]$ defined by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some } k, 1 \leq k \leq p \\ 0 & \text{otherwise.} \end{cases}$$

72

Properties of Boolean Matrices

Theorem 4. *If $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are Boolean matrices of compatible sizes, then*

1. (a) $\mathbf{A} \vee \mathbf{B} = \mathbf{B} \vee \mathbf{A}$.
 (b) $\mathbf{A} \wedge \mathbf{B} = \mathbf{B} \wedge \mathbf{A}$.
2. (a) $(\mathbf{A} \vee \mathbf{B}) \vee \mathbf{C} = \mathbf{A} \vee (\mathbf{B} \vee \mathbf{C})$.
 (b) $(\mathbf{A} \wedge \mathbf{B}) \wedge \mathbf{C} = \mathbf{A} \wedge (\mathbf{B} \wedge \mathbf{C})$.
3. (a) $\mathbf{A} \wedge (\mathbf{B} \vee \mathbf{C}) = (\mathbf{A} \wedge \mathbf{B}) \vee (\mathbf{A} \wedge \mathbf{C})$.
 (b) $\mathbf{A} \vee (\mathbf{B} \wedge \mathbf{C}) = (\mathbf{A} \vee \mathbf{B}) \wedge (\mathbf{A} \vee \mathbf{C})$.
4. $(\mathbf{A} \odot \mathbf{B}) \odot \mathbf{C} = \mathbf{A} \odot (\mathbf{B} \odot \mathbf{C})$.